Innovations in Surgical Training: Exploring the Role of Artificial Intelligence and Large Language Models (LLM)

Abstract

The landscape of surgical training is rapidly evolving with the advent of artificial intelligence (AI) and its integration into education and simulation. This manuscript aims to explore the potential applications and benefits of AI-assisted surgical training, particularly the use of large language models (LLMs), in enhancing communication, personalizing feedback, and promoting skill development. We discuss the advancements in simulation-based training, AI-driven assessment tools, video-based assessment systems, virtual reality (VR) and augmented reality (AR) platforms, and the potential role of LLMs in the transcription, translation, and summarization of feedback. Despite the promising opportunities presented by AI integration, several challenges must be addressed, including accuracy and reliability, ethical and privacy concerns, bias in AI models, integration with existing training systems, and training and adoption of AI-assisted tools. By proactively addressing these challenges and harnessing the potential of AI, the future of surgical training may be reshaped to provide a more comprehensive, safe, and effective learning experience for trainees, ultimately leading to better patient outcomes.